
Software Requirements Specification

for

furlink

Version 1.0

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Contents

CONTENTS	II
REVISIONS	II
1 INTRODUCTION	1
1.1 DOCUMENT PURPOSE	1
1.2 PRODUCT SCOPE	1
1.3 INTENDED AUDIENCE AND DOCUMENT OVERVIEW	1
1.4 DEFINITIONS, ACRONYMS AND ABBREVIATIONS	2
1.5 DOCUMENT CONVENTIONS	3
1.6 REFERENCES AND ACKNOWLEDGMENTS	4
2 OVERALL DESCRIPTION	6
2.1 PRODUCT OVERVIEW	6
2.2 PRODUCT FUNCTIONALITY	7
2.3 DESIGN AND IMPLEMENTATION CONSTRAINTS	7
2.4 ASSUMPTIONS AND DEPENDENCIES	8
3 SPECIFIC REQUIREMENTS	9
3.1 EXTERNAL INTERFACE REQUIREMENTS	9
3.2 FUNCTIONAL REQUIREMENTS	9
3.3 USE CASE MODEL	9
4 OTHER NON-FUNCTIONAL REQUIREMENTS	11
4.1 PERFORMANCE REQUIREMENTS	11
4.2 SAFETY AND SECURITY REQUIREMENTS	11
4.3 SOFTWARE QUALITY ATTRIBUTES	12
5 OTHER REQUIREMENTS	12
APPENDIX A – DATA DICTIONARY	13
APPENDIX B - GROUP LOG	14

Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Draft Type and Number	Full Name	Information about the revision. This table does not need to be filled in whenever a document is touched, only when the version is being upgraded.	00/00/00

1 Introduction

<TO DO: Please provide a brief introduction to your project and a brief overview of what the reader will find in this section.>

1.1 Document Purpose

<Identify the product whose software requirements are specified in this document, including the revision or release number. Describe the scope of the product that is covered by this SRS, particularly if this SRS describes only part of the system or a single subsystem.

TO DO: Write 1-2 paragraphs describing the purpose of this document as explained above.>

The purpose of this Software Requirements Specification (SRS) document is to clearly define and describe all functional and non-functional requirements for the furlink, version 1.0. This serves as a detailed guide for stakeholders including developers, project managers, testers, and end-users. Ensuring that all sides share a common understanding of the system's goals, constraints, and expected behavior.

1.2 Product Scope

<Provide a short description of the software being specified and its purpose, including relevant benefits, objectives, and goals.

This proposed project aims to develop a website platform to connect dog or cat owners with dog or cat grooming service providers in Makati City, creating more efficient access to pet grooming services. Meanwhile, it also allows service providers to have a wider and a more inclusive reach to their potential customers and organization of their booking system. As a result, this proposed project aims to increase the economic growth through amplifying the pet service industry, making this proposal aligned with SDG number 8.

The project includes dog or cat owners in Makati City that seeks new service options and are open to digital solutions. Another inclusion are small to medium-sized dog or cat grooming service providers based in Makati City that are not located in well-known areas

1.3 Intended Audience and Document Overview

<Describe the different types of reader that the document is intended for, such as developers, project managers, marketing staff, users, testers, and documentation writers (In your case it would probably be the "client" and the professor). Describe what the rest of this SRS contains and how it is organized. Suggest a sequence for reading the document, beginning with the overview sections and proceeding through the sections that are most pertinent to each reader type.>

It's essential to acknowledge the target beneficiaries of this project, especially people who are involved in the pet care industry. More specifically, this project is significant to the following groups of people:

- Pet Owners – This proposed project helps pet owners to have efficient access to reliable and nearby pet service providers. Improve user experience and provide convenience in booking pet services through the online platform.
- Pet Service Providers – This proposed project helps pet service providers reach a wider market than their usual local area. Increase visibility and improve organization of booking and scheduling through the system's digitalization.

- Local Communities – This proposed project helps the local community in making quality pet services accessible, improving their standards of pet care in the area, and boosting pet related services through gaining business activity on the platform.
- Academic Researchers in IT and Business –
- This study will help in providing data source and data driven insights for researches in digital pet service booking, user experience, and the pet service market.

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1.4 Definitions, Acronyms and Abbreviations

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.

TO DO: Please provide a list of all abbreviations and acronyms used in this document sorted in alphabetical order.>

A

AI (Artificial Intelligence) – Technology that enables machines to perform tasks that typically require human intelligence, such as image generation or prediction.

API (Application Programming Interface) – A set of tools and protocols that allow different software components to communicate.

C

Client – The end user or organization for whom the system is being developed.

COMET (Collaborative Object Modeling and Architectural Design Method) – A model-driven software design methodology used to structure system architecture and object models.

Cloud-Based Database – A database hosted on remote servers accessed through the internet rather than using local or on-premise infrastructure.

D

Dashboard – A user interface that provides analytics, summaries, and performance indicators related to system activities.

Deliverables – Documents, modules, or outputs required for submission according to the PBL schedule.

Database – A structured collection of data stored and accessed electronically, in this project handled through Supabase.

F

Furlink – The name of the proposed web-based pet grooming service booking platform.

G

Grooming Preview (AI Grooming Tool) – A feature that generates an estimated visual representation of a pet's haircut using AI technology.

H

HTML (HyperText Markup Language) – Standard language used to structure and display content on the web.

L

Local Service Provider – A small or medium-sized pet grooming business within Makati City using the platform to reach customers.

M

Messaging Channel – An in-platform communication feature that allows pet owners and service providers to exchange messages.

MVP (Minimum Viable Product) – An early version of the system that includes essential features for initial adoption and testing.

P

PBL (Project-Based Learning) – The instructional approach governing the project timeline, milestones, and deliverables.

Pet Owner – The end user looking to book grooming services for their dog or cat.

Platform – The Furlink website where users interact, book, and manage grooming services.

S

SDG (Sustainable Development Goal) – UN-defined global goals; the project aligns with SDG No. 8, focusing on economic growth.

Service Listing – A feature where grooming providers publish available services and pricing.

Service Provider / Pet Service Provider – A business offering dog or cat grooming services through the platform.

Supabase – A cloud-based backend platform providing database, authentication, and storage services.

U

UI (User Interface) – The visual elements users interact with in the platform.

UML (Unified Modeling Language) – A standardized modeling language used to represent software structure, behavior, and architecture in diagrams.

User – Any individual interacting with the system, including pet owners and service providers.

User Registration – The account creation process required before accessing platform features.

W

Web Platform – The online system where Furlink's services are delivered and accessed via a browser.

1.5 Document Conventions

<In general this document follows the IEEE formatting requirements. Use Arial font size 11, or 12 throughout the document for text. Use italics for comments. Document text should be single spaced and maintain the 1" margins found in this template. For Section and Subsection titles please follow the template.

TO DO: Describe any standards or typographical conventions that were followed when writing this SRS, such as fonts or highlighting that have special significance. Sometimes, it is useful to divide this section to several sections, e.g., Formatting Conventions, Naming Conventions, etc.>

1.6 References and Acknowledgments

<List any other documents or Web addresses to which this SRS refers. These may include user interface style guides, contracts, standards, system requirements specifications, use case documents, or a vision and scope document. >

References

- [1] Statista, "Total dog population Philippines 2020-2029," Statista, Apr. 14, 2025. <https://www.statista.com/statistics/1366302/philippines-dog-population/>
- [2] C. de Lazo, "Philippines' Burgeoning Pet Culture is Shifting Lifestyles—and it's Creating 'Paw-Some' Opportunities," Vero, Apr. 11, 2024. <https://vero-asean.com/philippines-pet-culture-brand-opportunities/>
- [3] K. S. Tan, "7 Pet Hotels In Metro Manila For A Stress-Free Stay For Your Furbaby While You're Away," TheSmartLocal Philippines - Travel, Lifestyle, Culture & Language Guide, Mar. 09, 2023. <https://thesmartlocal.ph/pet-hotels/>
- [4] G. Baron, "PVMA laments veterinarian shortage," Daily Tribune, Sep. 27, 2024. <https://tribune.net.ph/2024/09/27/pvma-laments-veterinarian-shortage#:~:text=The%20Philippine%20Veterinary%20Medical%20Association%20%28PVMA%29%20has%20recently,roughly%20over%2014%2C000%20registered%20veterinarians%20in%20the%20Philippines.>
- [5] A. Herrick, "Attention Small businesses without a website: find the right solution now," Web Design Booth - Your Home For The World Wide Web, Feb. 05, 2025. <https://www.webdesignbooth.com/businesses-without-a-website/>
- [6] G. N. Team, "The pet industry in the Philippines - GlobalPETS," GlobalPETS, Feb. 12, 2025. <https://globalpetindustry.com/news/the-pet-industry-in-the-philippines/>
- [7] E. Klarence, M. Decena, and Delfin, "Factors Affecting the Usage of Online Booking Sites: Comparative Analysis on Agoda, Booking, and Hotels," Jun. 2022. Available: <https://ieomsociety.org/proceedings/2022orlando/188.pdf>
- [8] "The Philippines: Steadily growing demand in this up- and-coming market," GlobalPETS, Dec. 28, 2023. <https://globalpetindustry.com/article/philippines-steadily-growing-demand-and-coming-market/>
- [9] S. E. McDonald, J. Sweeney, L. Niestat, and C. Doherty, "Grooming-Related concerns among companion animals: Preliminary data on an overlooked topic and considerations for animals' access to Health-Related services," Frontiers in Veterinary Science, vol. 9, Feb. 2022, doi: 10.3389/fvets.2022.827348.
- [10] "Revolutionizing pet care industry: HIVO." <https://hivo.co/blog/revolutionizing-the-pet-care-industry-with-digital-asset-management>

[11] K. M. Udvarhelyi-Tóth, I. B. Iotchev, E. Kubinyi, and B. Turcsán, "Why do people choose a particular dog? A Mixed-Methods analysis of factors owners consider important when acquiring a dog, on a convenience sample of Austrian pet dog owners," *Animals*, vol. 14, no. 18, p. 2634, Sep. 2024, doi: 10.3390/ani14182634.

[12] D. Buhalis and J. Chan, "Traveling with pets: designing hospitality services for pet owners/parents and hotel guests," *International Journal of Contemporary Hospitality Management*, vol. 35, no. 12, pp. 4217–4237, Mar. 2023, doi: 10.1108/ijchm-10-2022-1192.

[13] K. Steeves, "What Exactly Is an Online Booking System?," *Checkfront*, Oct. 01, 2020. <https://www.checkfront.com/blog/what-is-an-online-booking-system/>

[14] "What is an online booking system and how does it work?," *Zoho*, 2025.

<https://www.zoho.com/bookings/explore/what-is-online-booking-system.html>

[15] A. Parnaby, "Online Booking Systems: The Pros and Cons from the Customer's Perspective," *Grow your service business and get more bookings - SimplyBook.me*, Feb. 27, 2023. <https://news.simplybook.me/online-booking-systems-the-pros-and-cons-from-the-customers-perspective/>

[16] "What is an online booking system and how does it work?," *Zoho*, 2025.

<https://www.zoho.com/bookings/explore/what-is-online-booking-system.html>

[17] X. Chen and S. Cao, "How We Store and Process Millions of Orders Daily," *Grab Tech*, Aug. 15, 2022. <https://engineering.grab.com/how-we-store-millions-orders>

[18] "Agoda Intelligence - Agoda Partner Hub," Jun. 24, 2022.

<https://partnerhub.agoda.com/hotel-solutions/agoda-intelligence/>

[19] "pandapurpose 2023: Delivering with purpose to local communities across Asia | foodpanda news and blogs," *foodpanda*, 2023. <https://careers.foodpanda.com/blog-post/2024-7/pandapurpose-2023-delivering-with-purpose>

[20] "MANAGING THE DEVELOPMENT OF DIGITAL MARKETPLACES IN ASIA," 2021. Available: <https://www.adb.org/sites/default/files/publication/761016/managing-digital-marketplaces-asia.pdf> development-

[21] "Salesforce," *Salesforce*, 2024. <https://www.salesforce.com/commerce/conversational-commerce/>

[22] M. Lončarić, "Everything you need to know about conversational commerce," *Infobip*, Oct. 27, 2023. <https://www.infobip.com/blog/everything-you-need-to-know-about-conversational-commerce>

[23] "What Is Conversational Commerce?," *Bloomreach*, Sep. 19, 2024.

<https://www.bloomreach.com/en/blog/conversational-commerce>

[24] "Appointments - PetHealthHub," *Pethealthhubph.com*, 2022. <https://pethealthhubph.com/appointments>

- [25] "PetBacker Philippines," Petbacker.com, 2019. <https://www.petbacker.com/d/philippines>
- [26] author, "Best Practices for Tax Preparation Marketing," Textellent, Aug. 22, 2022. <https://textellent.com/blog/5-ways-to-use-sms-marketing-services-in-pet-care-businesses/>

2 Overall Description

2.1 Product Overview

<Describe the context and origin of the product being specified in this SRS. For example, state whether this product is a follow-on member of a product family, a replacement for certain existing systems, or a new, self-contained product. If the SRS defines a component of a larger system, relate the requirements of the larger system to the functionality of this software and identify interfaces between the two. In this part, make sure to include a simple diagram that shows the major components of the overall system, subsystem interconnections, and external interface. In this section it is crucial that you will be creative and provide as much information as possible.

TO DO: Provide at least one paragraph describing product perspective. Provide a general diagram that will illustrate how your product interacts with the environment and in what context it is being used. This is not a formal diagram, but rather something that is used to illustrate the product at a high level.>

As a startup, furlink aims to modernize the way pet grooming services are booked and managed in the Philippines. Unlike platforms such as Agoda or Airbnb—which focus on property and hotel bookings—furlink specializes in connecting dog or cat owners with pet grooming service providers through a streamlined digital platform.

furlink offers a hassle-free experience for both customers and service providers. It enables grooming businesses to advertise their services while allowing pet owners to discover, schedule, and manage appointments in one place. Additionally, the platform integrates an innovative AI-powered grooming preview tool, giving users a visual reference of potential pet haircut styles before booking—bridging the gap between customer expectations and actual grooming outcomes.

Designed as a local-first startup solution, furlink initially targets grooming services within select cities, with plans to expand as adoption grows.

2.2 Product Functionality

<Summarize the major functions the product must perform or must let the user perform. Details will be provided in Section 3, so only a high level summary is needed here. These can be at the level given in the project description.>

This project will include the following features:

1. User registration and login: Allows both dog or cat owners and service providers to create an account and login to their account.
2. Service listing: Allows service providers to post their services on the platform.
3. Booking: Enables dog or cat owners to book pet services on the platform.
4. AI pet haircut generator: Enables dog or cat owners to upload a photo of their pet and receive an AI-generated preview for their desired haircut style.
5. Messaging channel: Allows dog or cat owners and pet service providers to communicate with each other.
6. Pet service provider recommendation: Provides dog or cat owners recommended pet service providers based on their activities.
7. Dashboard: Allows pet service providers to track bookings, revenue, and customer insights.
8. Feedback form: Allows dog or cat owners to submit their review to the service provider.

2.3 Design and Implementation Constraints

<Describe any items or issues that will limit the options available to the developers. These might include: hardware limitations (timing requirements, memory requirements); interfaces to other applications; specific technologies, tools, and databases to be used; parallel operations; language requirements; communications protocols; security considerations; design conventions or programming standards (for example, if the customer's organization will be responsible for maintaining the delivered software). You can be creative here to some degree. At a minimum, you need to identify that you must use the COMET method for software design and the UML modeling language. Make sure you provide references for both.>

The development of furlink system is put through several constraints that will limit the options that are available to develop and design furlink. First, the team must follow the PBL-mandated schedule for the submission of all project deliverables which gives a limited time for development. This limited development timeframe affects the level of system refinement, iteration cycles, and the complexity of features that can be implemented within the project period. Second, the system will use Supabase a cloud-based backend database with no support for any on-premise servers. This requires for the system to solely rely on Supabase's authentication, storage, and other database services.

This project will exclude the following features:

1. Booking of pet care services with insurance.
2. Other pet services such as veterinary care, pet supplies, etc.
3. Pet transportation services
4. The system does not process any payments. It does not use in-app payment through third-party gateways.
5. In-depth financial reporting tools.

2.4 Assumptions and Dependencies

<List any assumed factors (as opposed to known facts) that could affect the requirements stated in the SRS. These could include third-party or commercial components that you plan to use, issues around the development or operating environment, or constraints. The project could be affected if these assumptions are incorrect, are not shared, or change. Also identify any dependencies the project has on external factors, such as software components that you intend to reuse from another project.

TO DO: Provide a short list of some major assumptions that might significantly affect your design.>

Assumptions

- If there is a potential client that does not have a full working system.
- If there is a potential client, they will fully fund the financial requirements of the project.
- There will be one pet grooming service provider that will be onboard.

Constraints

- Limited financial resources.
- Heavily relies on open-source and free-tier tools.
- Limited time given.
- The MVP aims to be completed by February 21, 2026.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

<Describe the logical characteristics of each interface between the software product and the users. For your project, you only need to be concerned with the main thermostat (not the mobile app) and can use the graphic from the project description as the basis for your user interface..

TO DO: Provide the graphic for the thermostat user interface and provide a basic description as to how users will interact (e.g. touch screen, menus, etc.).>

3.1.2 Hardware Interfaces

<Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the nature of the data and control interactions between the software and the hardware. You are not required to specify what protocols you will be using to communicate with the hardware, but it will be usually included in this part as well.

TO DO: Please provide a short description of the different hardware interfaces. This can simply be a list of the devices you must interact with at this point. For sensors, you can assume that each has a basic "read" interface to get the current values. Each sensor in this system uses English units.>

3.1.3 Software Interfaces

<Describe the connections between this product and other specific software components (in your case, just the mobile app that can send commands).>

3.2 Functional Requirements

< Functional requirements capture the intended behavior of the system. This behavior may be expressed as services, tasks or functions the system is required to perform. This section is the direct continuation of section 2.2 where you have specified the general functional requirements. Here, you should list in detail the different product functions.

3.2.1 F1: The system shall ...

3.2.2 <Functional Requirement or Feature #2>

...

3.3 Use Case Model

TO DO: Provide a use case diagram that will encapsulate the entire system and all actors.

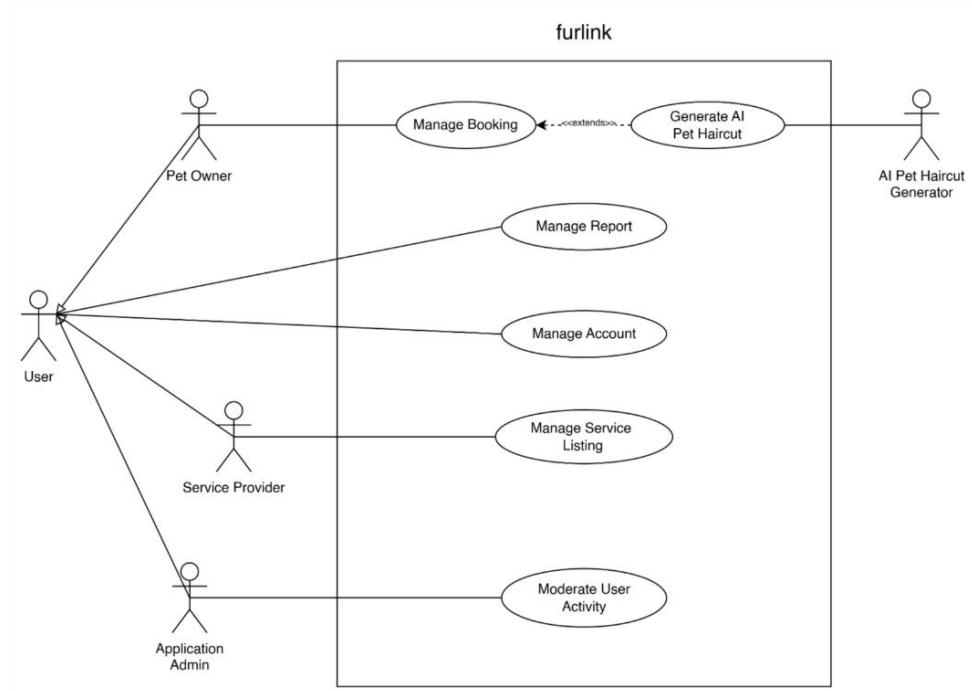


Figure 1 Use Case Diagram

3.3.1 Use Case #1 (use case name and unique identifier – e.g. U1)

TO DO: Provide a specification for each use case diagram

Author – Identify team member who wrote this use case

Purpose - What is the basic objective of the use-case. What is it trying to achieve?

Requirements Traceability – Identify all requirements traced to this use case

Priority - What is the priority. Low, Medium, High. Importance of this use case being completed and functioning properly when system is depolyed

Preconditions - Any condition that must be satisfied before the use case begins

Post conditions - The conditions that will be satisfied after the use case successfully completes

Actors – Actors (human, system, devices, etc.) that trigger the use case to execute or provide input to the use case

Extends – If this is an extension use case, identify which use case(s) it extends

Flow of Events

1. Basic Flow - flow of events normally executed in the use-case
2. Alternative Flow - a secondary flow of events due to infrequent conditions
3. Exceptions - Exceptions that may happen during the execution of the use case

Includes (other use case IDs)

Notes/Issues - Any relevant notes or issues that need to be resolved

3.3.2 Use Case #2

...

4 Other Non-functional Requirements

4.1 Performance Requirements

<If there are performance requirements for the product under various circumstances, state them here and explain their rationale, to help the developers understand the intent and make suitable design choices. Specify the timing relationships for real time systems. Make such requirements as specific as possible. You may need to state performance requirements for individual functional requirements or features.

TODO: Provide performance requirements based on the information you collected from the client/professor. For example, you can say "P1. The secondary heater will be engaged if the desired temperature is not reached within 10 seconds">

4.2 Safety and Security Requirements

<Specify those requirements that are concerned with possible loss, damage, or harm that could result from the use of the product. Define any safeguards or actions that must be taken, as well as

actions that must be prevented. Refer to any external policies or regulations that state safety issues that affect the product's design or use. Define any safety certifications that must be satisfied. Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product. Define any user identity authentication requirements.

TODO:

- Provide safety/security requirements based on your interview with the client - again you may need to be somewhat creative here. At the least, you should have some security for the mobile connection.

4.3 Software Quality Attributes

<Specify any additional quality characteristics for the product that will be important to either the customers or the developers. Some to consider are: adaptability, availability, correctness, flexibility, interoperability, maintainability, portability, reliability, reusability, robustness, testability, and usability. Write these to be specific, quantitative, and verifiable when possible. At the least, clarify the relative preferences for various attributes, such as ease of use over ease of learning.

TODO: Use subsections (e.g., 4.3.1 Reliability, 4.3.2 Adaptability, etc...) provide requirements related to the different software quality attributes. Base the information you include in these subsections on the material you have learned in the class. Make sure, that you do not just write "This software shall be maintainable..." Indicate how you plan to achieve it, & etc...Do not forget to include such attributes as the design for change (e.g. adapting for different sensors and heating/AC units, etc.). Please note that you need to include **at least 2** quality attributes. You can Google for examples that may pertain to your system.>

5 Other Requirements

<This section is **Optional**. Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A – Data Dictionary/ERD

<Data dictionary is used to track all the different variables, states and functional requirements that you described in your document. Make sure to include the complete list of all constants, state variables (and their possible states), inputs and outputs in a table. In the table, include the description of these items as well as all related operations and requirements.>

Appendix B - Group Log

<Please include here all the minutes from your group meetings, your group activities, and any other relevant information that will assist in determining the effort put forth to produce this document>

Appendix C – Test Plan/Test Cases